

MRP Title: Electric Vehicle Charging Demand Forecasting and Peak Load Risk Analysis

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Abstract:

This research proposes a machine learning framework to:

1. Analyze electric vehicle charging station utilization patterns using the Electric Vehicle Charging Station Availability Tracking dataset.
2. Forecast short-term charging demand using statistical, machine learning, and deep learning models, including ARIMA, SARIMA, Random Forest, XGBoost, LSTM, and GRU.
3. Identify peak-load risk periods and factors influencing charging station demand through exploratory data analysis and predictive modeling.

The framework integrates charging station operational data, temporal features, traffic conditions, weather information, and pricing variables to improve forecasting performance. Expected outcomes include accurate charging demand prediction, identification of high-demand periods, and practical recommendations for charging station management, infrastructure planning, and future electric vehicle adoption.

Research Questions and Objectives

The increasing adoption of electric vehicles has created a growing demand for reliable and efficient charging infrastructure. As charging station utilization continues to increase, forecasting charging demand has become important for charging network operators, infrastructure planners, and utility providers. Understanding charging demand patterns can help improve resource allocation, reduce congestion, and support future infrastructure development. This research aims to analyze electric vehicle charging demand patterns and evaluate forecasting approaches that can support operational decision-making

The study is guided by the following research questions:

Research Question 1: What charging demand patterns exist across different hours of the day and days of the week at public electric vehicle charging stations?

Research Question 2: How effectively can statistical, machine learning, and deep learning models forecast short-term electric vehicle charging demand?

Research Question 3: Which time periods are associated with the highest charging demand and peak-load risk?

Research Question 4: Which forecasting model provides the best predictive performance based on evaluation metrics such as RMSE and MAE?

To address these research questions, the study pursues the following objectives:

1. To perform exploratory data analysis and identify temporal, operational, and geographic patterns in electric vehicle charging demand.

2. To develop forecasting models using ARIMA, SARIMA, Random Forest, XGBoost, LSTM, and GRU algorithms.
3. To compare the forecasting performance of statistical, machine learning, and deep learning approaches using standardized evaluation metrics.
4. To identify peak-demand periods and charging station congestion risk using historical utilization data.
5. To provide practical recommendations for charging station operators and infrastructure planners based on forecasting results and demand analysis.

Literature Review

This literature review synthesizes seven research studies related to electric vehicle charging demand forecasting, time-series prediction, machine learning, and deep learning applications. The selected studies provide the theoretical and methodological foundation for this research by examining statistical forecasting models, machine learning techniques, deep learning architectures, and recent advancements in electric vehicle charging demand prediction. The review also highlights the strengths and limitations of existing approaches and identifies research gaps that motivate the current study. By analyzing previous work, this review establishes the rationale for comparing statistical, machine learning, and deep learning models for forecasting electric vehicle charging demand and identifying peak-load risk periods.

Description and the Summary of Papers

Box, Jenkins, Reinsel, and Ljung (2015), in their book **Time Series Analysis: Forecasting and Control**, introduced the **ARIMA** forecasting model, which has been widely used for time-series prediction. The model uses historical observations to identify patterns and forecast future values. Due to its simplicity and strong theoretical foundation, ARIMA has been applied in many areas such as electricity demand forecasting, transportation planning, and traffic flow prediction. The authors also discussed the **SARIMA** model, which extends ARIMA by considering seasonal patterns in the data. Since electric vehicle charging demand often follows daily and weekly patterns, these models are useful for forecasting charging station utilization. However, both ARIMA and SARIMA mainly focus on linear relationships and may not perform well when charging demand is affected by multiple external factors such as weather conditions, traffic congestion, and charging prices. The current project uses ARIMA and SARIMA as **baseline models** and compares them with machine learning and deep learning approaches to determine which method performs better for charging demand forecasting.

Hochreiter and Schmidhuber (1997) introduced the **Long Short-Term Memory (LSTM)** network, which became one of the most important developments in deep learning for sequential data analysis. Their study addressed the **vanishing gradient problem** found in traditional recurrent neural networks by introducing memory cells and gating mechanisms. This allowed the model to learn long-term dependencies and capture patterns over longer periods of time. Because electric vehicle charging demand often depends on historical charging behavior, LSTM models have become popular for demand forecasting applications. The main advantage of LSTM is its ability to learn complex temporal relationships automatically without requiring extensive feature engineering. However, LSTM models require larger datasets, more computational resources, and longer training times compared to traditional statistical methods. The current project evaluates **LSTM** as one of the forecasting models and compares its performance with statistical and machine learning approaches.

Chen and Guestrin (2016) developed **XGBoost**, a machine learning framework that has gained significant popularity due to its strong predictive performance and computational efficiency. The study showed that XGBoost can effectively handle large datasets and capture nonlinear relationships among variables. Unlike traditional statistical models, XGBoost can learn complex interactions between different factors without requiring strict assumptions about the data. In the context of electric vehicle charging demand forecasting, XGBoost can utilize variables such as traffic conditions, weather information, pricing, and temporal features to improve prediction accuracy. However, XGBoost does not directly model sequential dependencies in the same way as deep learning models. The current project includes **XGBoost** as a representative **machine learning model** and investigates whether it can outperform statistical and deep learning models for short-term charging demand forecasting.

Gedipudi (2023) published the **Electric Vehicle Charging Station Availability Tracking Dataset** on Kaggle, which serves as the primary dataset for this research. The dataset contains **100,000 charging station observations** collected at 30-minute intervals from twelve charging stations located across seven major cities. It includes information related to charging station utilization, weather conditions, traffic congestion, pricing, and temporal attributes. The availability of these variables provides an opportunity to explore charging demand patterns and identify factors influencing station utilization. Since the dataset contains both operational and contextual information, it is suitable for forecasting applications and comparative model evaluation. The current project directly uses this dataset to perform exploratory data analysis and develop charging demand forecasting models.

Shanmuganathan, Victoire, Balraj, and Victoire (2022), in their study on electric vehicle charging demand prediction, applied **LSTM networks** to forecast charging demand patterns. Their results showed that deep learning models were capable of capturing complex temporal relationships and achieved better forecasting accuracy than several traditional approaches. The study highlighted the usefulness of LSTM networks in handling nonlinear charging behavior and changing usage patterns. However, the research mainly focused on deep learning methods and did not provide extensive comparisons with machine learning ensemble techniques. The current project extends this work by comparing LSTM with machine learning models such as **Random Forest** and **XGBoost** as well as statistical forecasting methods.

Yi, Liu, Wei, Chen, and Dai (2022) investigated electric vehicle charging demand forecasting using **deep learning techniques**. Their research demonstrated that deep learning models could effectively capture charging demand patterns and improve forecasting performance when compared to traditional forecasting approaches. The authors also emphasized the importance of considering external variables such as weather conditions and traffic information when developing forecasting models. Their findings support the use of advanced forecasting techniques for electric vehicle charging applications. While the study mainly focused on deep learning models, the current project expands the analysis by evaluating statistical, machine learning, and deep learning approaches under the same experimental conditions.

Abdel-Aziz (2026) proposed a **hybrid forecasting model** that combines **XGBoost** and **Bidirectional Long Short-Term Memory (BiLSTM)** networks for electric vehicle charging station load forecasting. The results showed that the hybrid model achieved better performance than individual forecasting models. By combining machine learning and deep learning techniques, the model was able to capture both feature relationships and temporal patterns within the data. The study suggests that different forecasting approaches have unique strengths and that combining them may improve forecasting accuracy. However, hybrid models are often more complex to implement and require additional computational resources. This work motivates the comparative framework adopted in the current project, where multiple forecasting models are evaluated to better understand their strengths and limitations before identifying the most suitable approach for electric vehicle charging demand forecasting.

Integrated Insight: What Has Been Done and What Remains

The reviewed literature shows that statistical, machine learning, and deep learning approaches have all been successfully applied to electric vehicle charging demand forecasting. Statistical models such as ARIMA and SARIMA remain popular because they are easy to interpret and computationally efficient. Machine learning models such as XGBoost have demonstrated strong performance in capturing nonlinear relationships between charging demand and external factors. Deep learning models, particularly LSTM-based architectures, have shown the ability to learn complex temporal patterns and improve forecasting accuracy when large amounts of data are available.

The literature also highlights the growing interest in combining different forecasting approaches. Studies such as Abdel-Aziz (2026) showed that hybrid models can achieve better performance by leveraging the strengths of both machine learning and deep learning techniques. In addition, several studies emphasized the importance of contextual variables such as weather conditions, traffic congestion, pricing information, and temporal features when forecasting charging demand. These findings suggest that charging demand is influenced by a combination of operational, environmental, and behavioural factors rather than time alone.

Despite the progress made in previous studies, several research gaps remain. First, many studies focus on a single forecasting technique and provide limited comparison across statistical, machine learning, and deep learning model families. Second, a large portion of the literature focuses mainly on forecasting accuracy while giving less attention to how forecast results can support operational decision-making. Third, relatively few studies combine charging demand forecasting with peak-load risk analysis that can help charging station operators identify periods of high congestion. Finally, many existing studies rely on limited datasets or specific geographic locations, which may reduce the generalizability of their findings.

To address these gaps, the current project evaluates six forecasting models from three different model families using a common dataset and evaluation framework. The study also combines demand forecasting with peak-load risk analysis to provide practical insights for charging station operators and infrastructure planners. By comparing multiple forecasting approaches under identical conditions, the research aims to identify the most suitable method for short-term electric vehicle charging demand forecasting and operational planning

Conclusion

The literature review demonstrates that significant progress has been made in electric vehicle charging demand forecasting through the use of statistical, machine learning, and deep learning techniques. Previous studies have shown that each forecasting approach offers unique advantages and can contribute to improving charging demand prediction. At the same time, the review highlights several opportunities for further research, particularly in the areas of comparative model evaluation, operational application of forecast outputs, and peak-load risk identification.

Building upon the strengths of previous studies and addressing the limitations identified in the literature, this project proposes a comparative forecasting framework using ARIMA, SARIMA, Random Forest, XGBoost, LSTM, and GRU models. The selected dataset provides detailed operational, temporal, and contextual information that supports both demand forecasting and peak-load analysis. The findings from this research are expected to contribute to a better understanding of charging demand behavior and provide practical recommendations for charging station management, infrastructure planning, and future electric vehicle adoption.

Dataset Overview

1. [Electric Vehicle Charging Station Availability Dataset](#)

- **Size:** 100,000 records at 30-minute intervals.
- **Duration:** Six months (July 1, 2025 to December 31, 2025).
- **Coverage:** 12 charging stations across 7 cities (Los Angeles, San Francisco, San Diego, Phoenix, Denver, Salt Lake City, Seattle).
- **Networks:** 6 distinct charging networks.
- **Features:**
 - **Station identifiers:** station_id, station_name, network, city, state, latitude, longitude.
 - **Utilization metrics:** ports_total, ports_available, ports_occupied, ports_out_of_service, utilization_rate (primary target variable, range 0.02-0.98).
 - **Temporal attributes:** timestamp, hour_of_day (0-23), day_of_week (0-6), month (7-12), is_weekend, is_peak_hour.
 - **Contextual variables:** weather_condition, temperature_f, precipitation_mm, traffic_congestion_index, current_price, pricing_type, gas_price_per_gallon, local_event.
 - **Operational metrics:** station_status, estimated_wait_time_mins, avg_session_duration_mins, location_type, charger_type, power_output_kw.
- **Use:** Train and evaluate six time-series forecasting models (ARIMA, SARIMA, XGBoost, Random Forest, LSTM, GRU) for short-term charging demand prediction and peak-load-risk identification.

Exploratory Data Analysis

Exploratory data analysis was conducted systematically to understand demand patterns, identify peak-risk periods, and inform feature engineering for predictive modeling. The analysis spans eight key visualizations covering target variable distribution, temporal patterns (hourly, daily, weekly), peak-load-risk windows, temporal stability, geographic variation, and feature correlations. Each analysis is presented with both visualization and quantitative interpretation. Together, these analyses directly address the research questions on temporal patterns and peak demand periods, while informing model design choices for the comparative forecasting framework.

Target Variable Distribution

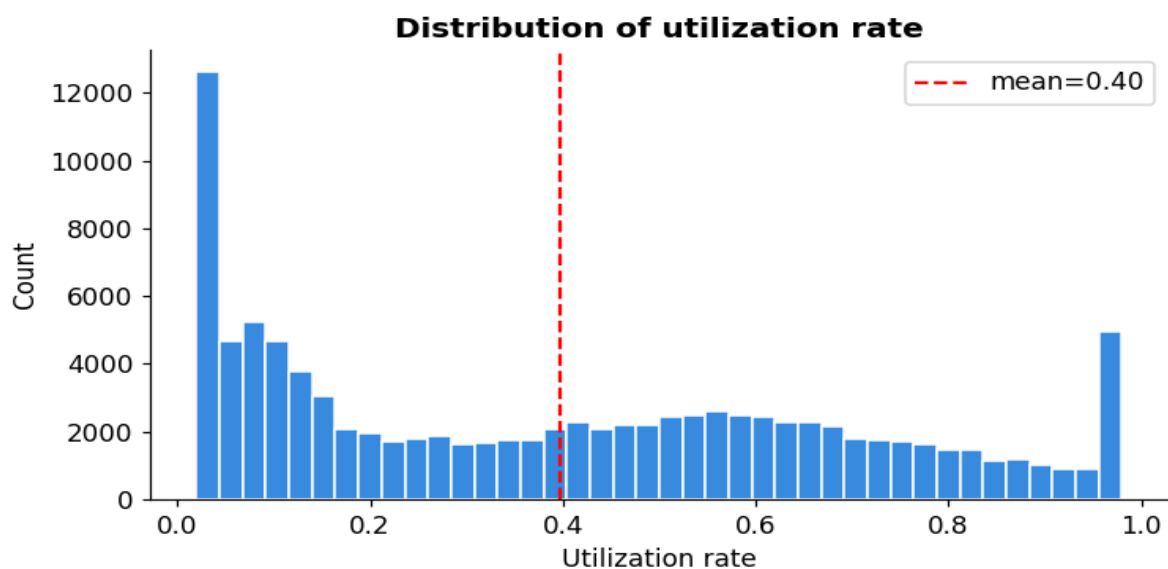


Figure 1: Distribution of station utilization rate across 100,000 records. Approximately bimodal pattern with peaks at moderate utilization (0.3-0.5) reflects diverse operational regimes.

The utilization rate distribution shows mean 0.40, standard deviation 0.18, ranging from 0.02 (overnight minimum) to 0.98 (peak congestion). The distribution is approximately bimodal—one mode at typical operational levels (0.3-0.5) and a secondary mode at peak periods (0.6-0.7). Approximately 10% of observations represent peak-load conditions (utilization above 0.70), identifying a meaningful proportion of congestion events that motivate the peak-load-risk analysis. This bimodality directly supports the project's methodological choice to separate routine operations from peak periods.

Hourly Demand Patterns

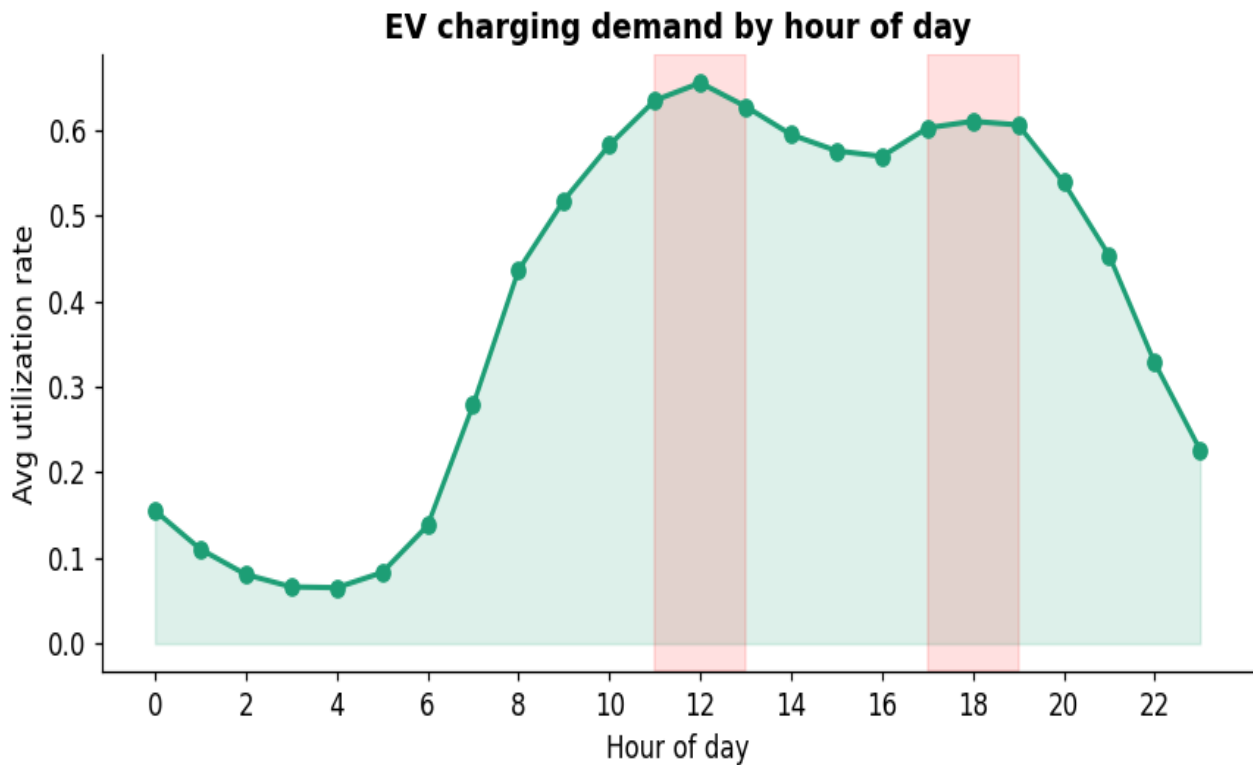


Figure 2: Hourly demand pattern with double-peak structure. Noon peak (0.66) and evening peak (0.52); minimum at 4 AM (0.07).

Hourly analysis reveals a pronounced double-peak pattern in charging demand: utilization climbs through morning hours, peaks at noon (0.66), drops slightly through afternoon, rises to a secondary peak at 5-7 PM (0.52), and declines sharply overnight to a minimum at 4 AM (0.07). Peak utilization is approximately 9 times higher than minimum utilization, indicating substantial infrastructure stress during peak periods and underutilization during overnight hours. This 9-fold variation has direct implications for dynamic pricing, capacity planning, and demand response programs.

Day-of-Week Variation

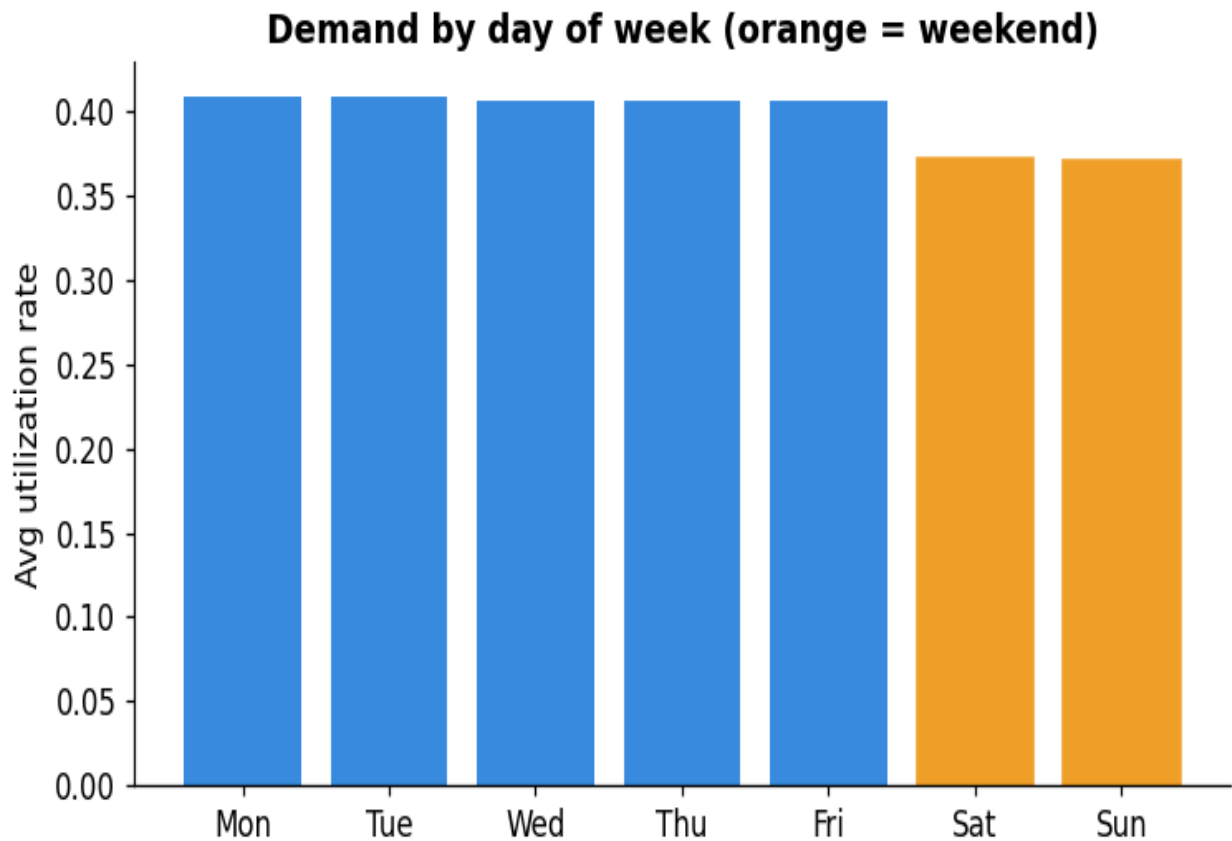


Figure 3: Average utilization by day of week. Weekdays consistently higher than weekends (orange). Weekday mean: 0.41; Weekend mean: 0.37.

Day-of-week analysis shows weekday demand (mean 0.41) approximately 10% higher than weekend demand (mean 0.37). Tuesday, Wednesday, and Thursday show highest weekday demand, while Sunday shows lowest demand overall—consistent with commuting-driven charging behavior dominating overall patterns. This finding directly supports the inclusion of day-of-week as a predictive feature in machine learning and deep learning models.

Weekday vs Weekend Hourly Profiles

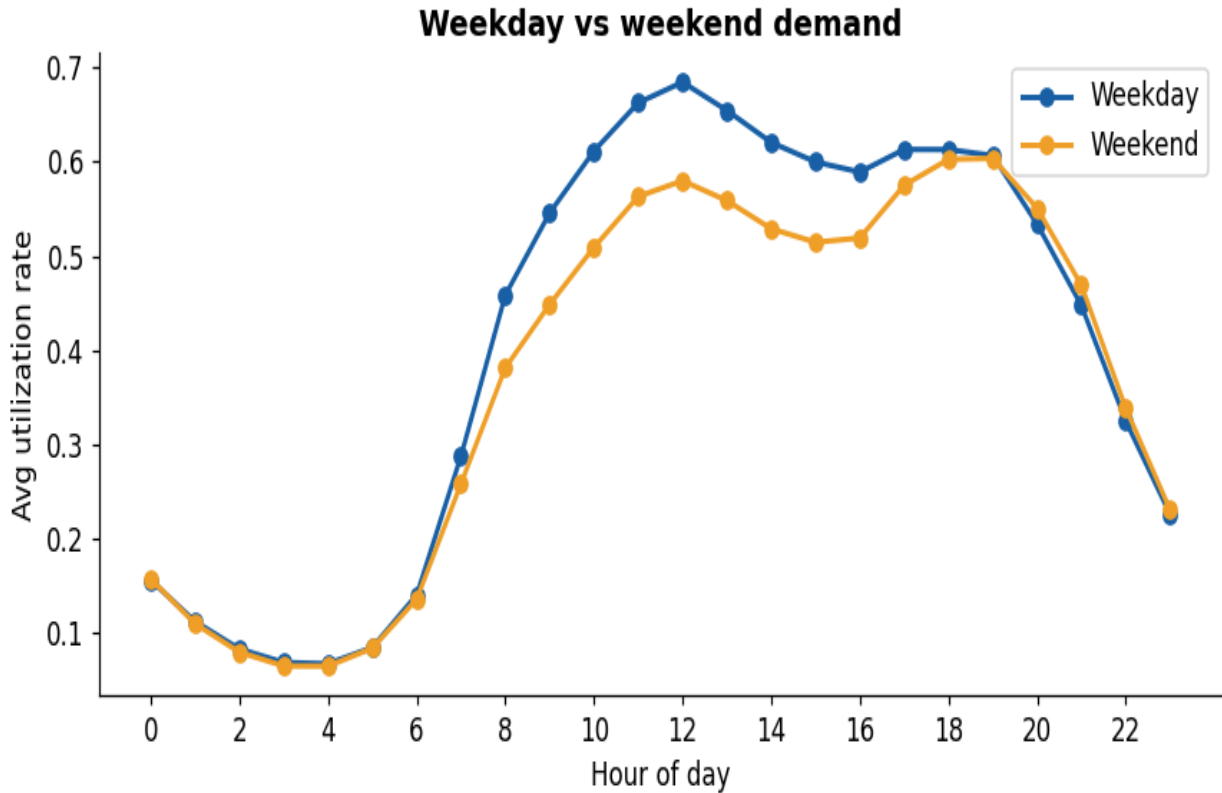


Figure 4: Weekday vs weekend hourly profiles. Both exhibit double-peak structure but weekdays show consistently higher absolute levels.

Comparison of weekday and weekend hourly profiles shows both regimes exhibit the same fundamental double-peak structure, but weekdays consistently show higher absolute levels throughout the day. The morning peak is more pronounced on weekdays (consistent with commuting), while the evening peak occurs at similar times in both regimes. This suggests forecasting models can use weekday/weekend as a binary feature without requiring fundamentally different model structures for the two regimes.

Peak-Load-Risk Heatmap

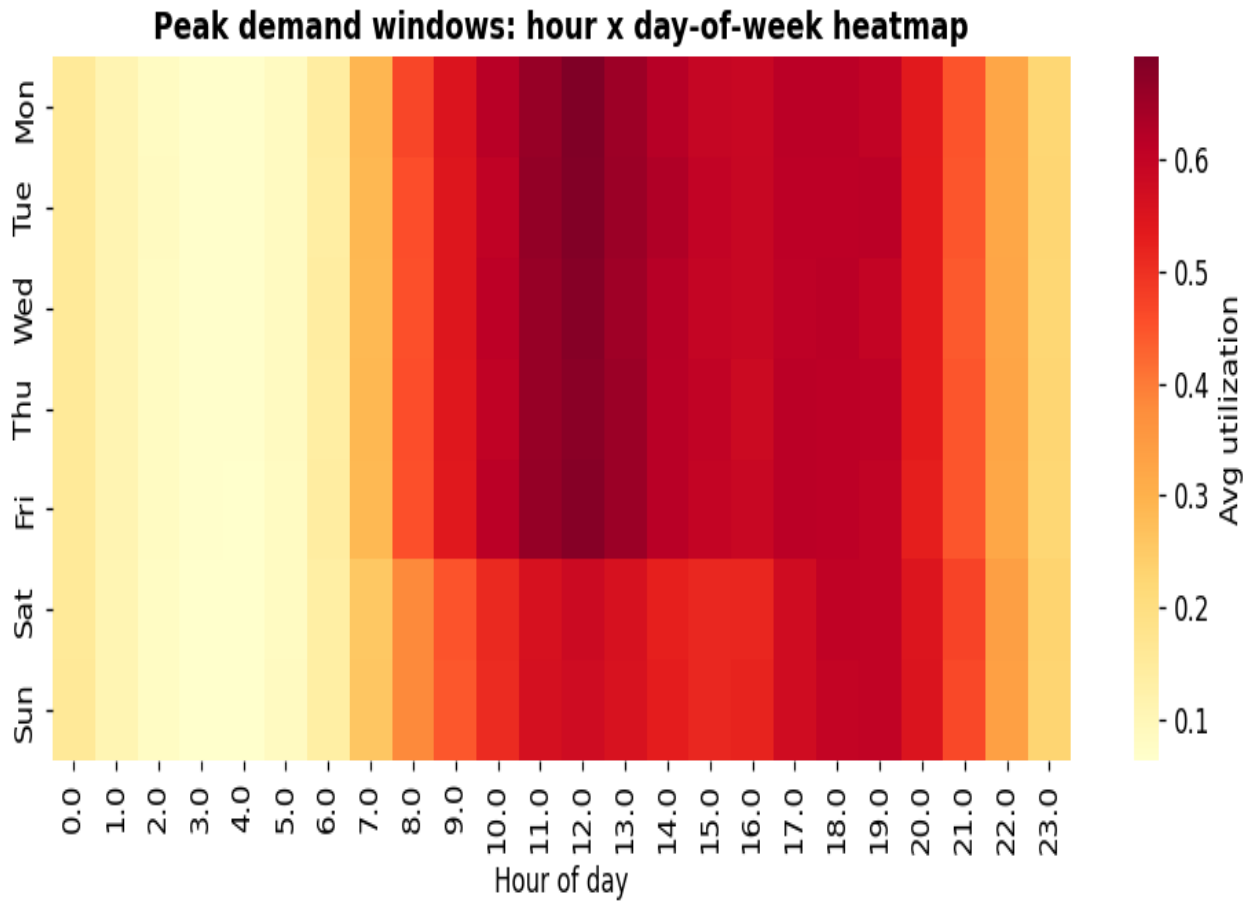


Figure 5: Peak-load-risk heatmap (hour x day). Darker red cells indicate higher congestion risk. Weekday 11AM-7PM shows maximum risk.

The peak-load-risk heatmap identifies the highest-risk congestion windows: weekday late-morning through early-evening (11 AM to 7 PM), with Tuesday through Thursday afternoons (noon to 6 PM) showing the most intense demand (average utilization exceeding 0.50). These periods represent critical intervals when infrastructure is most stressed and operators face the greatest risk of supply shortage. Overnight hours (10 PM to 6 AM) show consistently low demand across all days, suggesting opportunities for off-peak pricing incentives and demand-shifting programs.

Temporal Stability

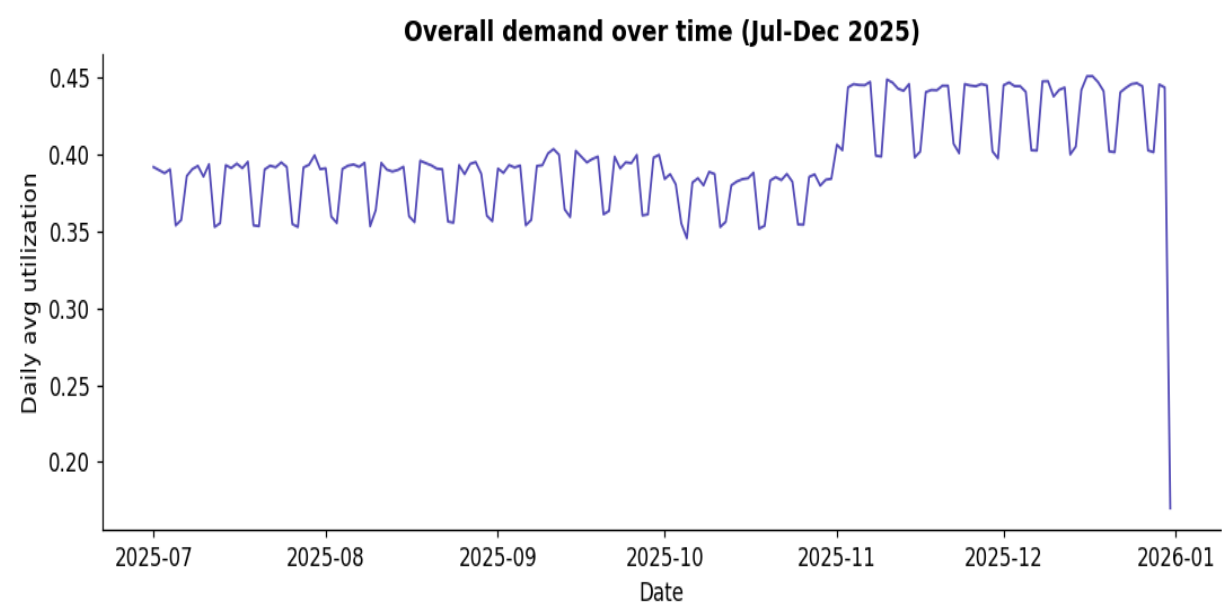


Figure 6: Daily average utilization across six months (July-December 2025). Demand is stable with no significant trend drift, supporting forecasting feasibility.

Time-series analysis reveals stable daily average utilization across the six-month observation period with no significant trend. This temporal stability suggests the underlying demand process is approximately stationary, that past patterns predict future demand, and that standard forecasting approaches are appropriate. This validates the use of historical data for training the six comparison models without concern about structural breaks or regime changes during the study period.

Geographic Variation



Figure 7: Average utilization by city. Los Angeles highest (0.58); Phoenix lowest (0.32), reflecting regional electric vehicle adoption variation.

Geographic analysis shows substantial variation across cities. Los Angeles exhibits the highest utilization (0.58) while Phoenix shows the lowest (0.32)—a nearly 2-fold difference reflecting regional differences in electric vehicle adoption rates, charging infrastructure density, urban density, and local policies. California cities consistently show higher demand, consistent with the state's aggressive electric vehicle policies. For modeling, this suggests that incorporating city or regional indicators may improve model performance by capturing systematic location-specific demand differences.

Feature Correlations

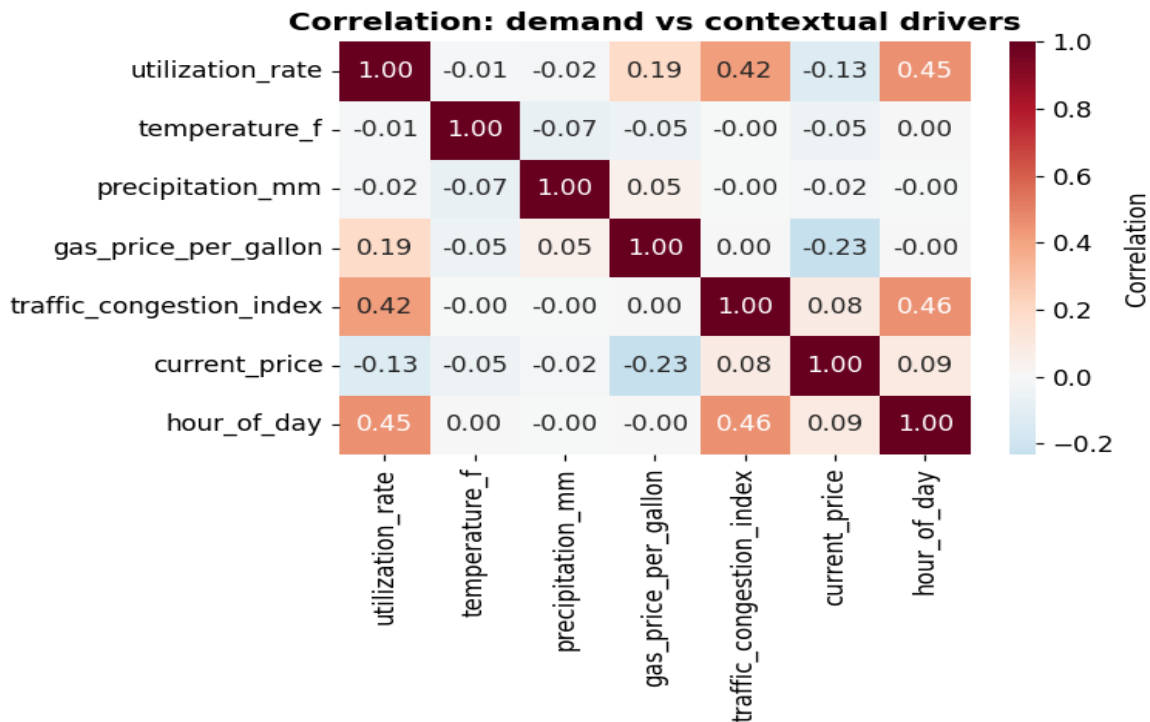


Figure 8: Correlation matrix. Hour of day strongest (+0.455), traffic congestion second (+0.416). Price shows weak negative correlation (-0.132).

Correlation analysis identifies hour of day ($r = +0.455$) as the strongest predictor of utilization, followed by traffic congestion ($r = +0.416$). Gas prices show weak positive correlation ($r = +0.189$), while charging price shows weak negative correlation ($r = -0.132$) indicating modest price sensitivity. These findings inform feature engineering: temporal features (hour, day) and contextual features (traffic, weather) will be incorporated as predictors in the machine learning and deep learning models. The relatively strong correlation with hour of day and traffic supports the use of these features in all model families.

Summary of EDA Findings

The exploratory analysis reveals six key insights that directly inform the modeling methodology: (1) Utilization demand follows a clear, stable double-peak pattern with peaks at noon (0.66) and evening hours (0.52); (2) Weekday demand is consistently approximately 10% higher than weekend demand, indicating commuting-driven behavior; (3) The peak-load-risk window spans weekday 11 AM to 7 PM, with Tuesday through Thursday afternoons showing maximum congestion risk; (4) Demand is stable across the six-month observation window with no significant trend drift, validating standard forecasting approaches; (5) Geographic variation is substantial with nearly 2-fold differences between cities; (6) Hour of day ($r = +0.455$) and traffic congestion ($r = +0.416$) emerge as the strongest predictors of utilization rate.

Methodology

This research project proposes a three-phase comparative forecasting framework to predict short-term electric vehicle charging demand and identify peak load risk periods at public charging stations. The pipeline integrates classical statistical models, modern machine learning ensemble methods, and deep learning recurrent neural networks under identical experimental conditions to enable fair head-to-head comparison across all three model families. The methodology adheres to established time-series forecasting practices including temporally-ordered data splitting, feature engineering, and rigorous evaluation using standard error metrics.

Below is the detailed description of each phase:

Phase 1: Statistical Forecasting Models

- **Dataset:** Daily-aggregated utilization rate from the Electric Vehicle Charging Station Availability Dataset (Gedipudi, 2023), with 184 daily observations across the six-month period.
- **Model:** ARIMA(1,1,1) as parsimonious baseline; SARIMA(1,1,1)x(1,1,1,7) extends ARIMA with weekly seasonality to capture observed weekly cycles.
- **Input:** Raw univariate time-series of daily average utilization (no feature engineering required).
- **Output:** Forecasted utilization rates for the test period with point estimates and confidence intervals
- **Justification:** Establishes an interpretable, computationally efficient baseline against which more complex models are compared. Statistical models are well-understood and serve as benchmarks for evaluating whether additional model complexity is justified.

Phase 2: Machine Learning Forecasting Models

- **Dataset:** Same daily-aggregated utilization data with engineered features (184 observations).
- **Models:** XGBoost (`n_estimators=100`, `max_depth=5`, `learning_rate=0.1`) for gradient boosting; Random Forest (`n_estimators=100`, `max_depth=10`) for ensemble averaging.
- **Feature Engineering:** Lag features (1, 2, 3, and 7 intervals prior), rolling statistics (7-interval mean and standard deviation), and temporal indicators (hour of day, day of week, weekend flag).
- **Input:** Feature matrix combining lag values, rolling statistics, and temporal context.
- **Output:** Forecasted utilization values plus feature importance rankings.
- **Justification:** Tests whether tree-based ensemble methods that capture non-linear feature interactions and contextual information can outperform purely temporal statistical models.

Phase 3: Deep Learning Forecasting Models

- **Dataset:** MinMax-scaled daily utilization sequences with 7-day lookback windows.
- **Models:** LSTM with 50 hidden units, 0.2 dropout, dense output layer; GRU with identical architecture using gated recurrent units instead of LSTM cells.
- **Training:** Adam optimizer with learning rate 0.001, 50 epochs, batch size 8, validation split 0.1.

- **Input:** Sequences of 7 consecutive past utilization values (lookback window) per prediction.
- **Output:** Forecasted utilization for the next time step; predictions are inverse-scaled back to original units.
- **Justification:** Tests whether deep learning recurrent networks that automatically learn temporal patterns can outperform feature-engineered statistical and machine learning approaches.

Evaluation Framework

- **Data Split:** Temporally-ordered 60/20/20 train/validation/test split preserving time-series structure.
- **Metrics:** Root Mean Squared Error (RMSE) as primary metric; Mean Absolute Error (MAE) as complementary metric.
- **Comparison:** All six models evaluated on identical test set; best model identified by lowest RMSE.
- **Peak-Load-Risk Analysis:** Identification of records in the top 10% utilization (90th percentile) and their distribution across hour-of-day and day-of-week, providing actionable operational insights.

Overall Architecture:

The system is designed as a three-phase comparative pipeline where six forecasting models from three families (statistical, machine learning, deep learning) are trained on the same dataset and evaluated using identical metrics on a held-out test set. The methodology ensures fair comparison through consistent experimental design, while the inclusion of peak-load-risk analysis translates forecast outputs into actionable operational insights for charging network operators. The pipeline is fully reproducible with all code available in the project GitHub repository.

Three-Tier Forecasting Pipeline for Electric Vehicle Charging Demand

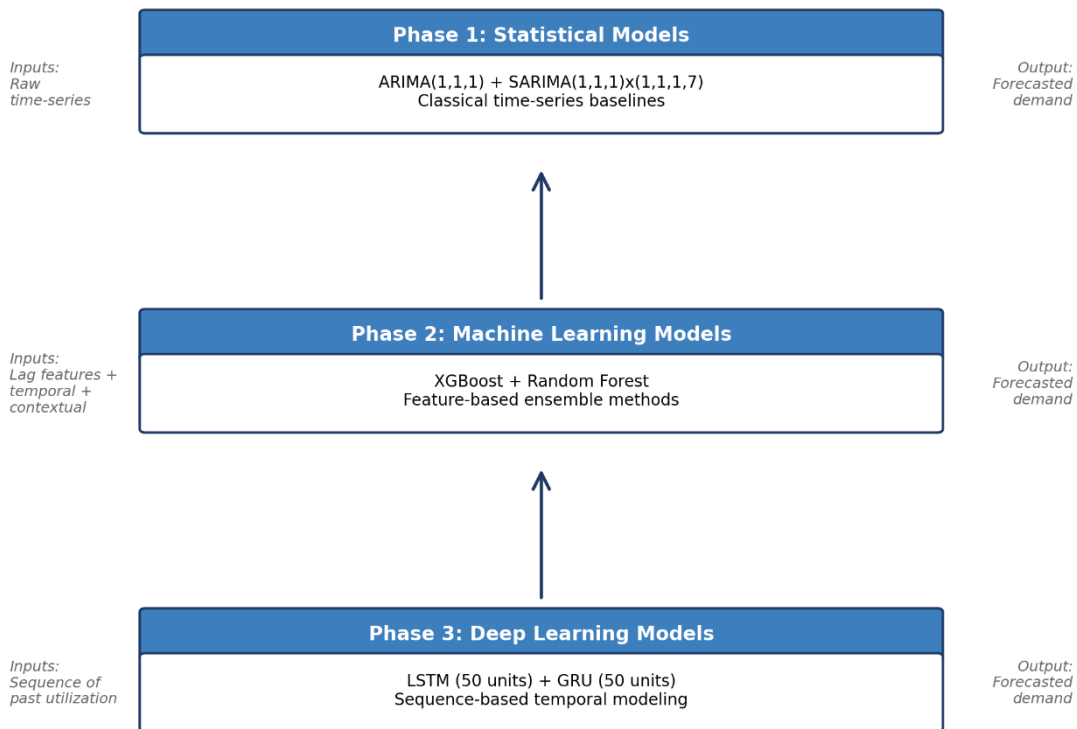


Figure: Overall Architecture of the Three-Phase Forecasting Pipeline

Github Repository link:

<https://github.com/karthikrajvenkatesh25/Electric-Vehicle-Charging-Demand-Forecasting-and-Peak-Load-Risk-Analysis>

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